

**ALLOWABLE STRESSES FOR CRACK
INITIATION**

Part: **4.7 – FATIGUE ALLOWABLE
STRESSES CALCULATION**

Chapter: **4710**

Title: **ALLOWABLE STRESSES FOR CRACK
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Revision: **0.1**

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1 INTRODUCTION

The term “fatigue allowables” can be understood as the material design values, applicable in initial stages of the airframe sizing process, when fatigue loads and spectra are not known in detail, which intend to prevent design flaws due to fatigue in those early moments, and are usually expressed as allowable stresses to compare with stress levels at Limit Load.

2 METHOD DESCRIPTION

The calculation of these fatigue allowables must necessarily be supported in simplifications and assumptions, being the resulting design values typically not really accurate but, typically, overly conservative. Suggested assumptions and calculation method are discussed in following lines.

Typical hypothesis

- Maximum fatigue stress not greater than 0.8 times that for Limit Loads
- Scatter factor (as using mean-values of S-N curves) of $SF = 5$
- Fatigue damage caused by one cycle per flight reaching the abovementioned maximum stress of 0.8 times LL with a fatigue stress ratio $R = -1$
- Typical design features/areas and associated stress concentration factors

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Kt_{net}	Description
1.5	Areas with stress concentration (effect of surface roughness)
2.5	Machined/folded radii, thickness changes, open holes with big edge distance...
3.0	Continuous riveted joints without big loads transfer
4.0	Fastened/bolted joints with high loads transfer

Table 1. Stress concentration factor for typical design features

Calculation process

- Calculate target life $N = SF \cdot DSG$
- Get associated maximum fatigue stress from the S-N curve of the material for the prescribed R and Kt values
- Transform that value into an allowable gross stress to compared with loads a LL level, using the expression $\sigma_{all} = \sigma_{max} \cdot (1/0.8) \cdot (1/SAF)$

Example

Inputs

- Structure made of aluminium alloy 7475T7351
- In areas with continuous riveted joints lightly loaded (eg. a frame flange fastened to a skin panel), thus applicable $Kt_{net} = 3$ and $SAF = 1.33$ (ratio gross to net-section area for fastener holes with edge distance equal to twice hole diameter)
- $DSG = 20000$ flights
- Fatigue stress ratio $R = -1$ and scatter factor $SF = 5$

Calculation

- $N = SF \cdot DSG = 5 \cdot 20000 = 100000$
- $\sigma_{max} = 81.5 \text{ MPa}$ (see following figure)
- $\sigma_{all} = \sigma_{max} \cdot (1/0.8) \cdot (1/SAF) = 81.5 \cdot (1/0.8) \cdot (1/1.33) = 76 \text{ MPa}$ (*)

(*) The Stress team in charge of the structure sizing may refer this design value to an allowable stress applicable at UL level, multiplying it by the typical Safety Factor (ratio UL/LL) of 1.5, resulting 115 MPa in this example.

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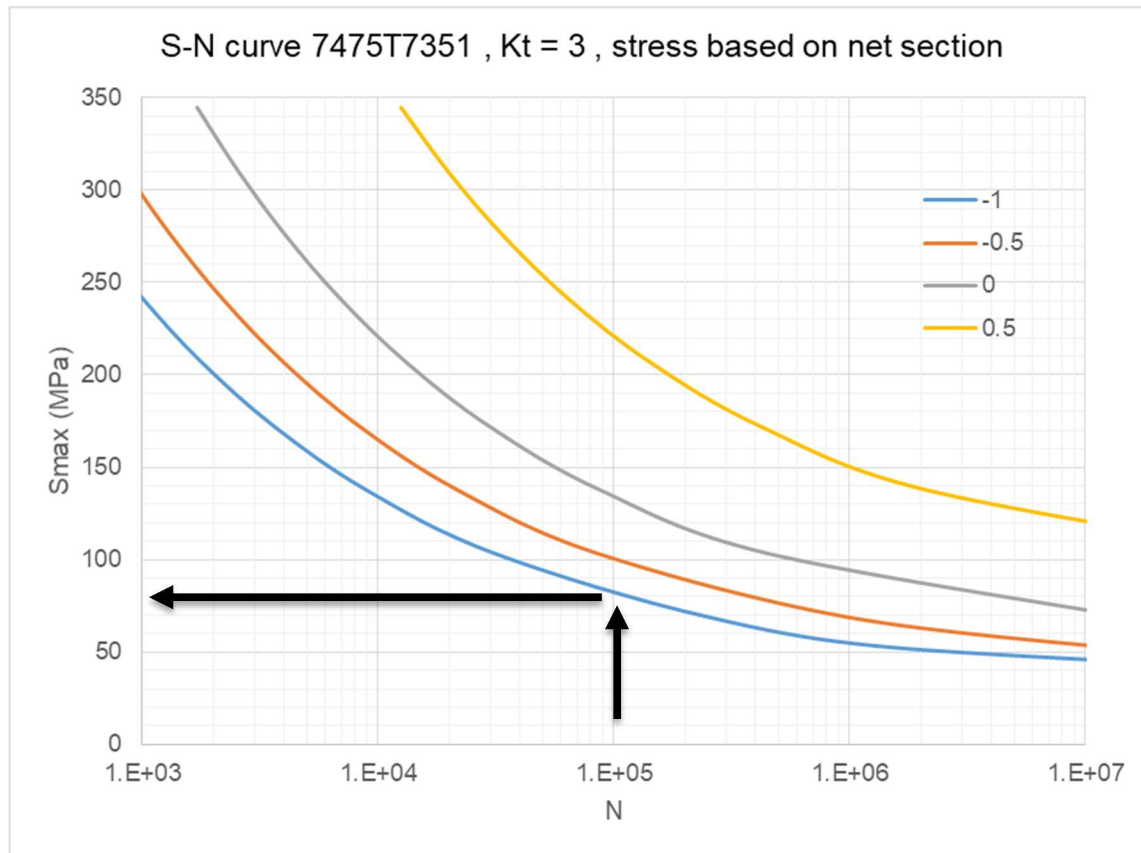


Figure 1. S-N curve of Al. 7475T7351 , from MMPDS-13 Figure 3.7.21.2.8(b) (plotted after conversion to SI units, using the reported Equivalent Stress equation)

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3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
<i>DSG</i>	- / hrs	Design Service Goal, in number of flights or in flight hours
<i>Kt</i>	-	Stress concentration factor, typ. given as maximum stress divided by the net section stress
<i>LL</i>	-	Limit Load
<i>R</i>	-	Fatigue stress ratio, i.e. is the minimum divided by the maximum stress in a cycle
<i>SAF</i>	-	Stress Adjustment Factor, i.e. ratio between net and gross stresses
<i>SF</i>	-	Fatigue Scatter Factor, or life reduction factor, is the statistically derived divisor applied to fatigue life to account for the variation in fatigue performance
<i>UL</i>	-	Ultimate Load
σ_{min}	MPa	Minimum stress in a fatigue cycle
σ_{max}	MPa	Maximum stress in a fatigue cycle

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[BMI] MMPDS	13	Metallic Materials Properties Development and Standardization
Ref. 2	MM-BM-ADF-04003	A	Admisibles de Fatiga

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
ALFA	??	Airbus DS software for calculating the fatigue allowables of metallic structures

Table 2. Software programs referenced in this chapter

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RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub-chapters affected
0.1 06/03/2023	Gustavo Baños	First issue	All pages

Table 3. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Stress Analysis (TEPAA)	José Efraín Mirón Rubio DD/MM/2023	Holger Hickethier DD/MM/2023
Stress Methods (TEPAP-TL4)	Gustavo Baños DD/MM/2023	-

Table 4. Approval signatures table for last revision of this chapter